

GURU PRABHAKAR

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Summary

Release Engineer with over 4 years of experience in connected systems, CI/CD pipelines, cloud platforms, and software release management. Strong focus on automation, secure software delivery, and distributed system testing. Interested in applying AI driven approaches and security testing methods to enhance verification pipelines and software quality in automotive compute platforms.

Technical Skills

Operating Systems: Windows, Linux | **Programming & Scripting:** Python, PowerShell, Bash, Groovy | **Test Automation & System Testing:** Pytest, Robot Framework, Hardware-in-the-Loop (HiL), Vector CANoe, Android Debug Bridge (adb), Trace & Log Analysis | **Automotive Networks & Protocols:** CAN/CAN-FD, Automotive Ethernet, VW Body Application Protocol (BAP) | **Digital Key & Platform:** CCC Mobile Device Key (MDK), MQB evo, Connectivity Module (ConMod) | **Vehicle-Cloud Integration:** MQTT, REST/HTTP APIs, TCP/IP | **Security:** TLS/mTLS, OAuth 2.0 | **CI/CD, DevSecOps & Release Management:** Jenkins, Azure Pipelines, GitHub/Bitbucket, GitHub Actions, Azure DevOps, Shift-Left Security, SAST, Key Vault, Container Scanning, JFrog Artifactory | **Cloud, Containers & Infrastructure:** Docker, Kubernetes/AKS, Infrastructure as Code (IaC) with Terraform, Azure Resource Manager (ARM) | **Monitoring & Observability:** Prometheus, Grafana | **AI-Assisted Testing & Development:** GitHub Copilot, M365 Copilot, Claude, Generative AI for Testing | **Project & Collaboration Tools:** Jira (test plans & dashboards), Confluence

Experience

Volkswagen Infotainment GmbH

10/2024 - Present

System Integration, Test and Release Engineer

Bochum, Germany

- Defined and executed test strategies for end-to-end system validation. Led end-to-end testing of the **Mobile Device Key (MDK, smartphone-as-key)** across the frontend, in-car modules, vehicle network, and backend; executed **integration** test runs with a **98% pass rate**.
- Orchestrated Hardware-in-the-Loop (HiL) test environments, integrating software and vehicle hardware components for end-to-end validation.
- Developed and maintained **Python based test automation** for system validation, health monitoring, and automated verification workflows, improving software quality and reducing manual test effort. Boosting automation test coverage from **0% to 80%** and reducing bench setup time by **90%**.
- Developed and tested a Python based application for automated data transformation and validation, implementing unit and functional tests to ensure reliability across end-to-end software workflows.
- Performed detailed **trace and log analysis** (Vector CANoe logs, MQTT traffic), resolving more than **18 cross-component defects** through multi-source timestamp and log correlation.
- Integrated automated test flows into **Jenkins CI/CD pipelines** (with nightly continuous testing) using CLI-driven execution flags, enabling unattended nightly system runs and faster feedback loops.
- Orchestrated **16 sprint-based releases** for the Vehicle Integration connectivity stack, managing promotions from develop to master branches, multi-repository synchronization, build requests, and quality gates; delivered **38 artifacts/week** across a 7-tier dependency chain.
- Developed **Python** based automation tool for the release pipeline to streamline and standardize delivery preparation.
- Managed sprint-based releases for the VI-ConMod software, handling Git branching/tagging, multi-repository synchronization, and artifact lifecycle in JFrog Artifactory with full traceability via Confluence checklists and Jira links.
- Performed OSS compliance for the connectivity software stack. Improved compliance automation by enhancing Python-based OSS tooling, including **SPDX normalization**, dual-direction **BSD-2/3-Clause validator**, license-file conformance checks, and a blacklist-candidate analyzer—reducing manual follow-up by **40%**.

TeamViewer

05/2023 - 07/2024

Software Release Engineer Working Student

Goeppingen, Germany

- Developed a **Python-based automation tool** to streamline the software release process, improving consistency and reducing manual effort.
- Integrated release workflows with **Jira** and **Confluence** via REST APIs and implemented **unit tests** for 30+ release modules to ensure reliability.

- Built cloud-based **CI/CD pipeline** using Azure DevOps and Azure CDN, cutting deployment time by **50%** and improving update availability.
- Provisioned Azure infrastructure using **ARM**, implementing Infrastructure as Code (IaC) practices for repeatable and secure deployments.
- Containerized release tools using **Docker** to ensure environment consistency across development and deployment stages.
- Optimized packaging and delivery workflows, achieving a **40% reduction** in software package size and a **20% reduction** in cloud operational costs.

Bosch Thermotechnik / Bosch Home Comfort

08/2022 - 01/2023

DevOps Engineer Intern

Wernau, Germany

- **Process Automation:** Automated embedded software release tasks using **Python** and the Bosch **QG-CLI** tool, enabling consistent and repeatable artifact retrieval from multiple sources.
- Developed Python scripts to integrate QG-CLI into a **Jenkins CI/CD pipeline**, achieving a **100% reduction in manual artifact fetch time** and improving release workflow reliability.
- Enhanced internal tooling by identifying defects, reporting issues, and proposing improvements to **QG-CLI**, contributing to more stable release automation.

Fraunhofer IOSB

08/2021 - 04/2022

Research Assistant

Ilmenau, Germany

- Conducted comparative analysis of existing network simulators and simulation frameworks.
- Simulation of different protocols for wireless communication using OMNeT++.
- Utilized the data from firmware to create a simulation model using OMNeT++ that emulates the behavior of network nodes within the RTLS.
- Implemented CI/CD pipeline to automatically test the software packages.

GE Healthcare

07/2018 - 06/2019

Manufacturing Process Trainee Engineer

Bengaluru, India

- Testing and Process Improvement: Implemented testing procedures for X-ray tube process equipment and supported product development.
- Technical expert for (High kV Generator) product manufacturing processes, worked with value-stream team on daily and long-term projects impacting production, effectively participated in cross-functional kaizen events by learning and then utilising Lean concepts.
- Supported development and evaluation of manufacturing processes for potential improvements and implementing change control procedures.

Education

Technical University of Ilmenau, Germany

10/2019 - 07/2024

Master of Science in Communications and Signal Processing

Ilmenau, Germany

Major Subjects: Communication Networks, Advanced Mobile Communication Networks, Cloud Computing.

Collaborative research project: Cloud Security and DevSecOps Infrastructure for Machine Learning based IoT threat Mitigation. Automated DevSecOps CI/CD pipelines with integrated SAST and dependency, vulnerability scans to containerize and deploy the ML based threat detection service on Azure Kubernetes Service (AKS). Automated cloud resource provisioning using Terraform (IaC) and established observability with Prometheus and Grafana.

Thesis: Investigation of Cloud-based Software Packaging and Release Management Systems at TeamViewer. Skills: Software Development Lifecycle, DevOps Practices, Microsoft Azure, Software Packaging and Deployment automation, CI/CD pipelines, Artifact Creation, Content Delivery Network (CDN).

Visvesvaraya Technological University - RNSIT, India

08/2014 - 06/2018

Bachelor of Engineering in Electronics and Communication

Bengaluru, India

Thesis: Pest Control System using Raspberry pi

Certifications

Microsoft Certified: Azure Fundamentals | Atlassian Agile Project Management Professional Certificate | DevOps Professional Certificate by PagerDuty and LinkedIn | Docker Foundations Professional Certificate | Career Essentials in GitHub Professional Certificate | ISTQB Certified Tester Foundation Level (CTFL) | ISTQB Certified Tester – Testing with Generative AI (CT-GenAI) (Training Completed — Certification Examination Scheduled: Aug 2026)

Language Skills

German: A2 (Beginner) | **English:** Fluent (Professional Proficiency)